Secure FSO Link Parts List

Part	Quantity	Purpose
BPW34 photodiode	2	Optical receiver
D6505I laser diode	2	Optical transmitter
OPA320 op-amp	2	Amplifies photodiode current
SN74AHCT125 buffer	2	Drives the transmitter switching signal
LMV7219 comparator	2	Converts the received signal to digital levels
SOT-23-5 to DIP adapter	5	Mounts for ICs
2N7000 MOSFET	5	Switch for laser current
22 ohm resistor	10	Limits ringing and switching current
FIT0203 perfboard	3	For receiver and transmitter circuits
HK-6.4 laser housing	2	Housing for laser
25mm piano-convex lens	1	Focuses beam onto photodiode
Red acrylic color filter	1	Ambient light filter
Black pipe or shroud	1	Ambient light blocker
82 kΩ	2	OPA320 transimpedance feedback resistor
100 kΩ	4	Creates the receiver’s half-supply reference voltage
1 kΩ	4	Current limiting around the comparator and controls
100 Ω	2	Laser-driver/current-path resistance
1 MΩ	2	Bias or discharge path where required
10 kΩ potentiometer	2	Adjustable comparator threshold
100 nF ceramic	8 to 10	One beside every IC supply pin/reference-voltage filtering
10 µF electrolytic	2 to 4	Bulk power filtering on each board
10 pF ceramic	2	OPA320 feedback compensation
22 pF ceramic	2	Alternative feedback value for tuning stability
47 pF ceramic	2	Additional tuning option if the amplifier oscillates